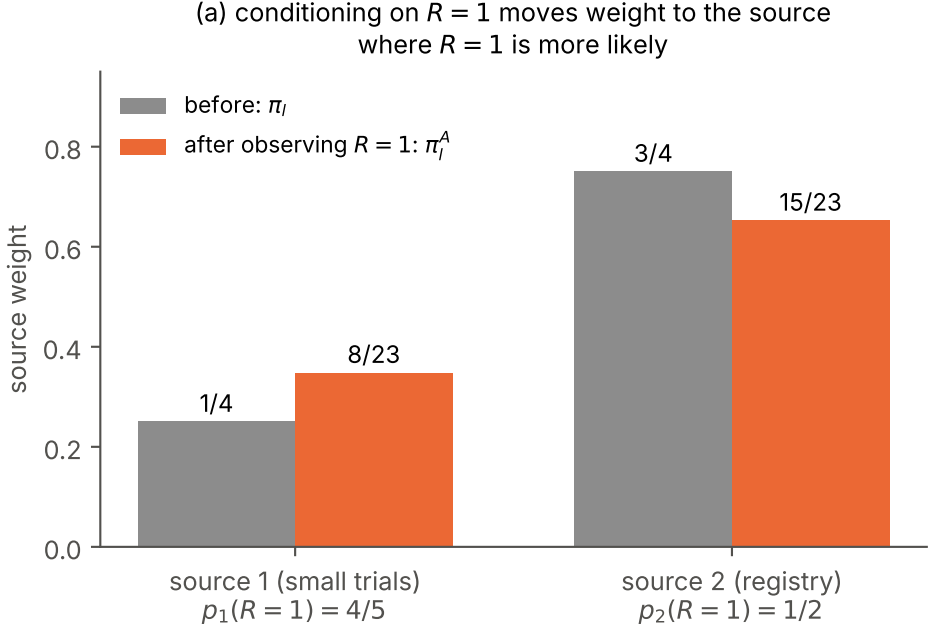
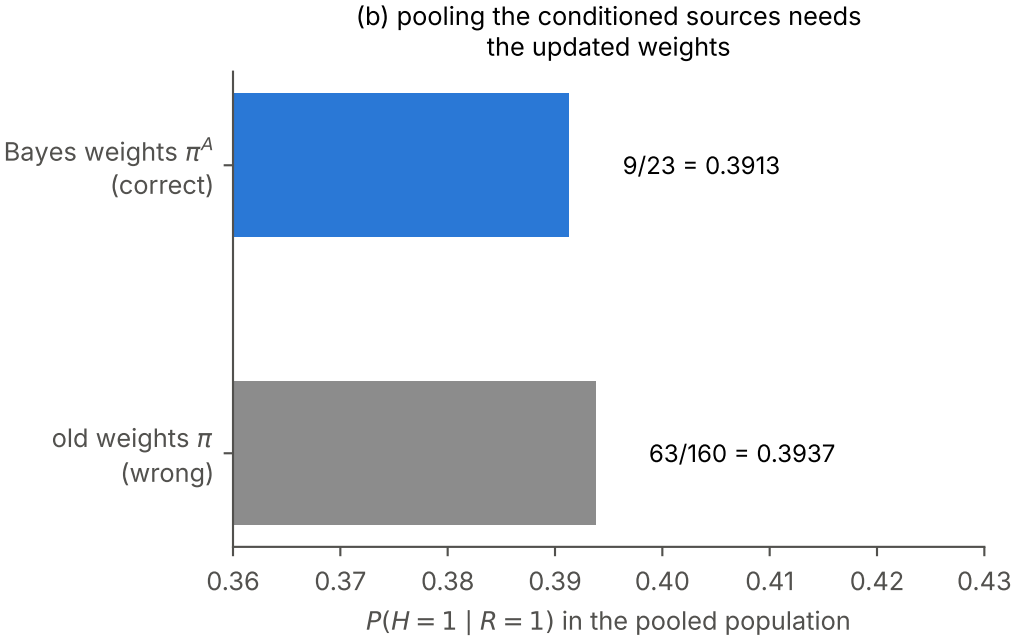

(a) conditioning on $R = 1$ moves weight to the source where $R = 1$ is more likely
before: $\pi_l$
after observing $R = 1$: $\pi_l^A$
source weight
0.0
0.2
0.4
0.6
0.8
1/4
8/23
3/4
15/23
source 1 (small trials)
$p_1(R = 1) = 4/5$
source 2 (registry)
$p_2(R = 1) = 1/2$


(b) pooling the conditioned sources needs the updated weights
Bayes weights $\pi^A$ (correct)
9/23 = 0.3913
old weights $\pi$ (wrong)
63/160 = 0.3937
0.36
0.37
0.38
0.39
0.40
0.41
0.42
0.43
$P(H = 1 \mid R = 1)$ in the pooled population